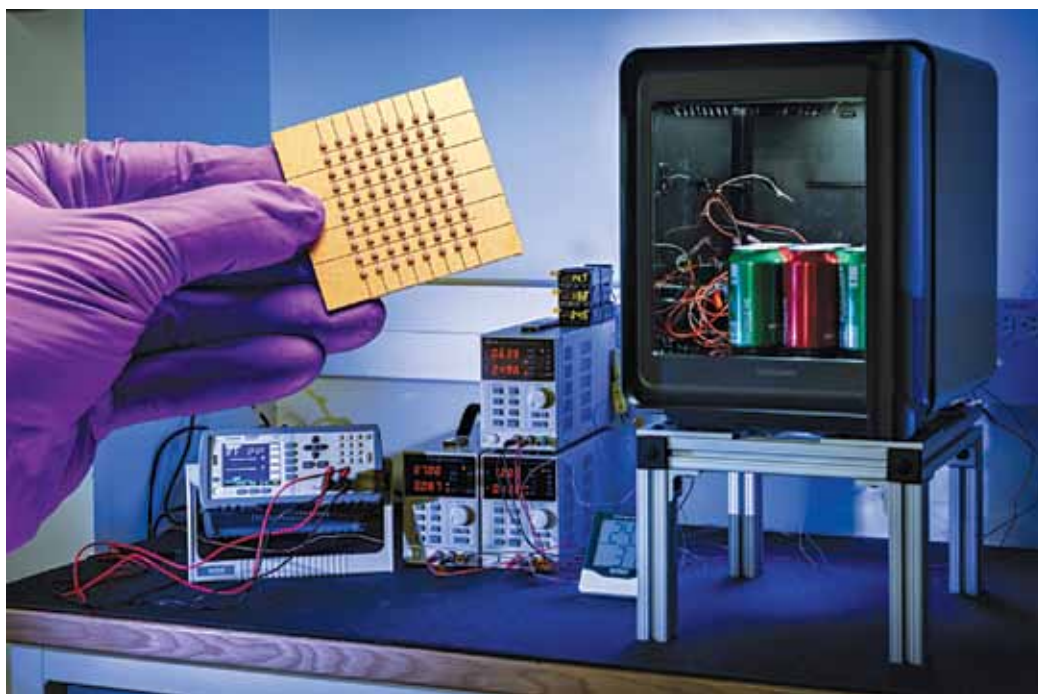


World's first high-performance Peltier refrigerator

Samsung Electronics and Johns Hopkins Applied Physics Laboratory have developed thin-film Peltier semiconductor

devices using advanced nano-engineering, which boosts cooling efficiency by nearly 75%, reduces material use to

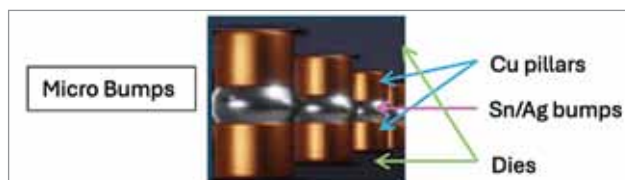


World's first high-performance Peltier refrigerator with nano thin-film tech represents leap toward refrigerant-free cooling (Credit: <https://news.samsung.com/global>)

1/1000 of the usual volumes, and simplifies manufacturing with miniaturised, lightweight designs. This enables high-performance Peltier refrigerators that cool faster and more precisely than traditional vapour compression systems, use no refrigerants, lower production costs, and support environmentally friendly mass production for industries like home appliances, medical devices, automotive electronics, and data centres.

Hydrogen plasma for better chip packaging and performance

Dr Robert F. Hicks and his team at Surfx Technologies developed an in situ atmospheric pressure hydrogen plasma method that uses ionised hydrogen plasma inside the bonding tool to remove metal oxides from copper by converting them into metal and water vapour, leaving no residue. This process avoids damage to micro-bumps, operates within chip temperature limits, supports interconnects, enables signal speeds exceeding 5GHz with low loss, and is compatible with various bonding types. As chips shrink and demand higher interconnect density for AI, 5G, and cloud



Micro-bump structure after thermocompression bonding (Credit: <https://www.semiconductorpkgnews.com>)

computing, this clean, residue-free method provides a reliable way to improve chip packaging and performance.